

Research Paper Kickoff

Welcome to what is arguably the most instructive part of the course, which is the research paper (and even more arguably, the most fun part). You will independently study energy-efficient computing (EEC) of a topic of your choice that hopefully you feel passionately about.

Goals:

There are several goals that we would like to achieve out of this effort:

- To broaden your horizons on the topic of energy-efficient computing beyond what is covered by the lectures.
- To acquire, apply or sharpen research skills.
- To acquire, apply or sharpen technical writing skills.
- To acquire, apply or sharpen presentation skills.

Type of Research Papers:

For this class, the research papers fall into one of the following categories:

- A breadth coverage of an idea/system/technology related to EEC.
- An in-depth analysis of an idea/system/technology related to EEC (encouraged).
- A project report detailing a project that you designed, carried out, and measured (highly-encouraged).

Scope:

You have a great deal of latitude to pick a topic of research that you are interested in, with the only condition being that it is related to energy-efficient computing. The topics can range from purely technical to even societal and beyond. You will need approval of the instructor before you start the research. Examples of topics that were research in previous years include:

Broad system issues:

- Energy-efficient cloud computing (includes scheduling, infrastructure, provisioning, quality of service, policies).
- Data center design and operation.
- Smart grids to feed the energy needs of computing.
- Communications systems.
- Mobile computing.
- Energy-aware storage systems.
- Benchmarking and measurements.
- Computing paradigms and energy (peer to peer, software as a service, serverless computing, service-oriented applications, etc.)

Application-level issues:

- Energy issues in cryptocurrencies and blockchain technologies.
- Efficient high-performance computing.
- Energy-aware programming.
- Energy-aware application design and implementation.

- Energy-aware distributed systems.
- Computer security interactions with energy.

Operating system issues:

- Operating system energy awareness and support for energy efficient computing.

Technology:

- Future technologies: Neuromorphic computing, quantum computing and bio computing.
- Beyond silicon: New (and old) material such as graphene, germanium, etc.
- Smart batteries.
- Hardware support for energy-efficient computing.

Societal issues:

- Computing's energy impact on climate change.
- Computing's energy cost impact on business profitability.

You are not limited to the topics listed above. However, you must in every case come up with a solid argument as to 1) why your topic of choice belongs to the course, 2) what you will cover beyond the lectures.

Logistics:

You will form groups of two students each. You will write a research proposal outlining the scope of your proposed research, why it relates to EEC, and you should show some preliminary references of papers or sources that you did some preliminary research on. The goal of the last step is to ensure that you have started on a reasonable path toward your end goal. The instructor will give you feedback on your proposal. Most of the time, your proposal will be accepted. However, your proposal will be rejected if 1) the instructor deems the topic unrelated to EEC, 2) the preliminary research does not appear relevant or sufficient, 3) the proposal is not clear, is written sloppily, or 4) the proposal is too ambitious to be carried out for an entire semester. Please refer to the class schedule for the deadline to submit the proposal and other scheduling issues.